Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.15	20.27	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.01	2.03	+0.6%
power (W), mean	630	642	+2.0%
power (W), peak	652	692	+6.0%
energy (Wh)	211	217	+2.6%
energy, parked workers still on at idle power (Wh)	211	217	+2.6%
CPU used (cores), mean	0.910	1.24	+36.2%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.038	0.052	+36.2%
energy per core-hour (Wh)	692	518	-25.1%
node-hours per core-hour	6.60	4.84	-26.6%
pending pods, pod-minutes	0.000	0.000	0
pending pods, peak	0.000	0.000	0
HPA replicas, mean	9.13	9.32	+2.0%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.533	0.517	-3.1%
requests timed	1,040	1,095	+5.3%
time over the response line (% of samples)	1.15	0.000	-100.0%
response time (ms), mean	156	98.04	-37.1%
response time (ms), median	128	84.90	-33.4%
response time (ms), 95th percentile	321	158	-50.7%
response time (ms), 99th percentile	524	192	-63.3%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.1986	5.5525	-22.9%	[-3.3118, -0.0204]	GAIN
kwh_per_core_hour	0.7521	0.5887	-21.7%	[-0.3205, -0.0012]	GAIN
utilisation	0.0374	0.0508	+35.7%	[+0.0008, +0.0272]	GAIN
pending_pod_minutes_per_hour	0.0000	0.0000	n/a	[+0.0000, +0.0000]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	1.5299	1.4877	-2.8%	[-2.3762, +2.2496]	EVEN, no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 0.695
- heat (thermal) per minute: 0.358 0.392 0.421 0.445 0.468 0.487 0.505 0.52 0.534 0.546 0.556 0.562 0.567 0.571 0.577 0.581 0.585 0.588 0.59 0.591
- nodes decided per minute: 6 5 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	617	621	6.00	5.25	0.000	0.000	0.575	0.683

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	6.00	620	622	6.00	6.00	0.000	0.000	0.657	0.705
2	6.00	6.00	620	628	6.00	7.00	0.000	0.000	0.652	0.872
3	6.00	6.00	623	636	6.00	8.50	0.000	0.000	0.716	1.07
4	6.00	6.00	635	650	9.25	10.00	0.000	0.000	1.05	1.43
5	6.00	6.00	634	646	10.00	10.00	0.000	0.000	1.05	1.35
6	6.00	6.00	634	651	10.00	10.00	0.000	0.000	1.01	1.47
7	6.00	6.00	647	670	10.00	10.00	0.000	0.000	1.38	2.00
8	6.00	6.00	646	680	10.00	10.00	0.000	0.000	1.34	2.24
9	6.00	6.00	650	670	10.00	10.00	0.000	0.000	1.45	1.97
10	6.00	6.00	646	669	10.00	10.00	0.000	0.000	1.31	1.96
11	6.00	6.00	616	623	10.00	10.00	0.000	0.000	0.565	0.739
12	6.00	6.00	620	623	10.00	10.00	0.000	0.000	0.659	0.746
13	6.00	6.00	625	628	10.00	10.00	0.000	0.000	0.777	0.879
14	6.00	6.00	635	650	10.00	10.00	0.000	0.000	1.04	1.46
15	6.00	6.00	632	649	10.00	10.00	0.000	0.000	0.974	1.37
16	6.00	6.00	634	654	10.00	10.00	0.000	0.000	1.03	1.55
17	6.00	6.00	630	633	10.00	10.00	0.000	0.000	0.912	0.992
18	6.00	6.00	618	625	10.00	10.00	0.000	0.000	0.589	0.789
19	6.00	6.00	619	623	10.00	10.00	0.000	0.000	0.622	0.734
20		6.00		626		10.00		0.000		0.805

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.